Response to Comments:

We would like to express our sincere gratitude to the editor and reviewers for their insightful and professional comments, which are greatly helpful for us to improve the quality of this manuscript. Notably, we have eliminated all the earlier blue revision marks in the revised draft. And newly marked-up version is newly highlighted in blue for straightforward reference and assessment.

Editor's Comment:

1.Consider the reviewers comments below if any.

Response: We have carefully considered the reviewer's comments, who mentioned that "The authors have adequately addressed the majority of my previous comments. Although minor aspects could still be further refined, the manuscript in its current form meets the requirements for publication." Accordingly, we have further polished and refined the minor details throughout the manuscript based on the editor's comments, to improve its overall quality.

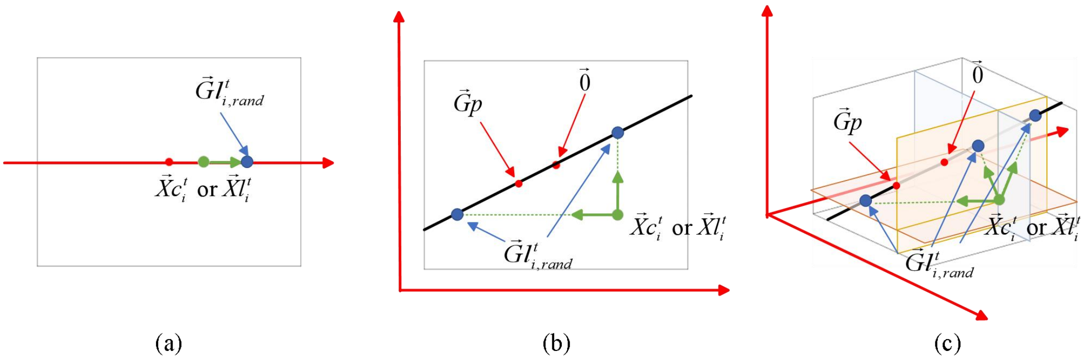
2.Use only institutional emails; other emails such as Yahoo or Gmail are not permitted.

Response: We have noted this rule clearly. All authors only use institutional email addresses without employing any personal Gmail, Yahoo or other non-institutional emails.

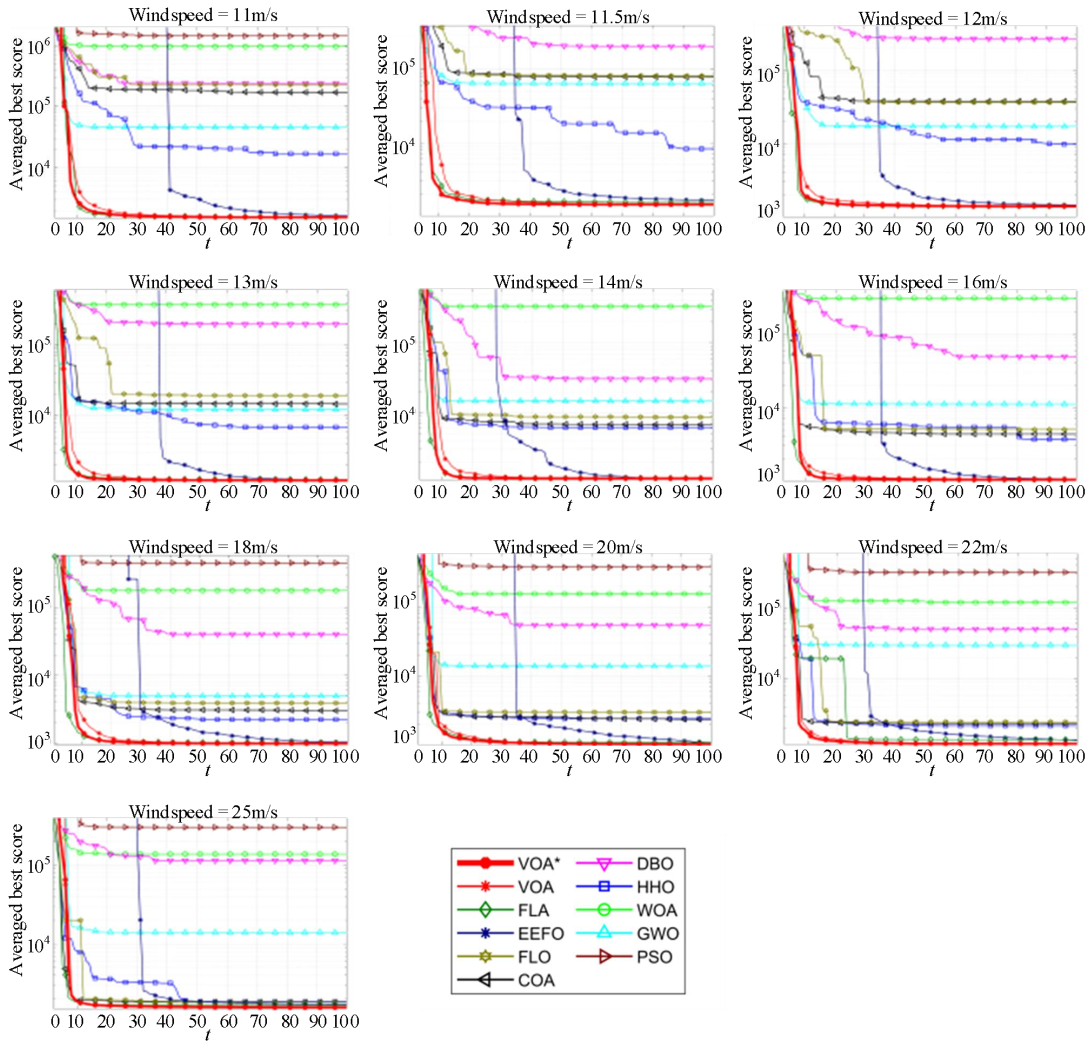
3.The paper figures should be of high quality, with readable text inside them.

Response: We have carefully revised all figures in the manuscript. All images have been updated to high-resolution versions with clear and readable text inside. Meanwhile, we have appropriately enlarged the size of each figure to further improve clarity and visual presentation.

Revised Fig.4:



Revised Fig.7:



4.Authors can have only 2 affiliations maximum per author; one affiliation is highly recommended.

Response: We have strictly followed this requirement. Each author is listed with no more than two affiliations, and we have simplified the academic affiliations as much as possible to adopt one single affiliation for most authors in accordance with the journal's recommendation.

5.Authors should include recent references that cover the state of the art; therefore, the authors are asked to add recent related work published in 2024 and 2025.

Response: Thanks for the constructive comment. As suggested, we have added multiple recent references from 2024 to 2025 in the Literature Review sections, which effectively supplement the state-of-the-art research background of this study. The specific additions are as follows.

Revised part:

Evolutionary-based algorithms are the first proposed MH algorithms, which are inspired by natural selection, heredity and evolution mechanisms in nature. Among these, genetic algorithm (GA)[11] is one of the most widely used. It generates new populations through gene selection, crossover, mutation and replacement, gradually converging toward an approximate optimal solution. In recent years, inspired by the reproduction and evolution of the elk herd, scholars have proposed a more efficient evolutionary-based algorithm called the elk herd optimizer (EHO)[12].

Swarm-based algorithms are the most popular and rapidly growing MH algorithms nowadays, which are inspired by the group behavior of biological populations in nature. The swarm-based algorithms could data back to the emergence of particle swarm optimization (PSO)[13] in 1995, which simulates the group behavior of birds and fish to find the optimal solution in many problems. Since then, many innovative swarm-based algorithms have developed quickly due to their intuitive understanding and strong performance, representative examples including bitterling fish optimization (BFO)[14] and gray langurs optimizer (GLO)[15].

Physics-based algorithms are driven by physics, chemistry, or mathematical models, involving mechanics, heat, electromagnetism, optics, atomic physics, chemical reactions, thermodynamics, mathematical functions, mathematical rules, and so on. They can rely on common formulas or rules to simplify the process of constructing optimization models, while also allowing users to quickly understand and utilize them. Inspired by sine and cosine functions, sine-cosine algorithm (SCA)[16] is one of the well-known physics-based algorithms. Although, many physics-based algorithms usually lack competitiveness due to their complex principles and rigid formulas, researchers still continuously explore and develop novel physics-based optimization algorithms in recent years, such as young's double-slit experiment optimizer (YDSE)[17] and fata morgana algorithm (FATA)[18].

Human-based algorithms simulate various aspects of human behavior, such as life, social interaction, movement, and creativity. Due to the inherent complexity of human behavior, human-based MH algorithms have long-term development potential. However, explaining complex human behaviors within these algorithms remains a significant challenge. For example, the hiking optimization algorithm (HOA)[19] is improved solely on Tobler's hiking function[20] proposed in 1979, which was not effective applied in Human-based algorithms until over forty years later. Even so, researchers continue to propose novel human-based metaheuristics and explore interpretable behavioral modeling in recent years. For instance, mother optimization algorithm (MOA)[21] is inspired by the caring behavior of mothers toward their children, while the user-follower interaction algorithm (UFIA)[22] originates from the observation and imitation behaviors of children toward family members.

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6.Removing old related work that is not closely related to the paper's scope is recommended as well.

Response: Thanks for your valuable suggestion. We have moved most basic experimental tests and corresponding discussions to the supplementary materials. Besides, after careful review, we confirm that the remaining content is closely relevant to the scope of this paper, and no further deletion or simplification is required.

7.A complete language check should be performed to correct all typos and language issues.

Response: Thanks for your valuable suggestion. We have thoroughly conducted a full language and grammar proofread throughout the entire manuscript. All typos, grammatical errors, and inappropriate expressions have been carefully revised and corrected to improve the readability and academic rigor of the paper, which are highlighted in blue in the marked-up version.